

Promotional text for Interleaves

A good friend had asked me to open for his blues-funk band at a bar during college. He thought it would be cool to start the evening with some classical guitar. All I remember from that evening is the terror seeping through my body as I stumbled from one note to the next in what was probably a completely incoherent performance of Antonio Lauro's Four Venezuelan Waltzes. I spiraled. I saw myself from the audience's perspective: a terrified charlatan who didn't know what he was doing on the stage. Most of my life I had been suffering from performance anxiety and struggled to understand why my performance was so erratic. I wanted so badly to make beautiful music that communicated something with the listener, but so many things seemed to get in the way. I chalked it up to stage fright and didn't know what to do about it.

One day I came across an online class from Juilliard, with Dr. Noah Kageyama. I don't remember the name of the class, but it was positioned to be about taming stage fright. Although it was a luxury I could barely afford, I decided to give it a go. Great decision!

At first I thought there was a bit of a bait and switch as it was mostly about practice techniques, and stage fright itself was only minimally covered. But I soon realized that how we practice is probably the most effective tool we have against stage fright. Things started turning around for me. Little by little I started making fewer mistakes, and more importantly, was able to stay "in the music" more often and for more time.

I started incorporating everything that I was learning into my students' lessons, and saw musical improvements and less anxiety in general around music making.

Since then, keeping up with the science around learning and performing music has been a part of my life. Constantly learning how to practice smarter, not longer has become more and more important to me. I look back at my younger self and wonder just what I thought I was accomplishing with those 6-hour practice sessions!

With all the money that has been poured into studies of the performance of professional athletes, pilots, surgeons, etc... It's fantastic that we are co-opting the

research for musicians. Basketball, tennis, golf, surgery — these are all activities where efficient, directed movement, and focus are key. Not so different for a musician.

One thing that many studies support is the value of interleaved practice. In an interleaved practice session, you mix up the material being practiced rather than playing through it sequentially. When we practice the same thing over and over, slowly getting better at it, we often end our practice session sounding pretty good, but the next day we are back closer to where we started than where we left off. This makes sense. *Of course* we are going to sound good the 15th time we play something in a row. Unfortunately, concerts don't work that way. We need to sound good the first time, so it seems logical that practicing passages as if it were the first time would be beneficial. Play a passage two or three times. Move on to another, then another, then back to the first. We are strengthening our recall, and getting better at hitting the passage correctly right from the start. Randomizing the order is even better: can we play it well the first time when we don't even know what's coming up? This prepares us for the unavoidable curveballs that will be thrown at us. A cell phone going off in the audience, an ensemble member playing the wrong note, us playing a wrong note and needing to recover, a memory of a traumatizing past performance, a noisy environment. It's not always a concert hall for most of us. Training ourselves to get right to the music at hand while chaos surrounds us is essential for secure performance.

Neuroscience has a lot to say about these practice methods. Repeating a passage over and over actually depletes the materials needed to myelinate (coat and insulate) the neural connection being formed, so after a certain point there is literally no benefit to continued repetitions. The brain needs to replenish the material and that takes time. During short breaks, our brains run through the actions we are learning and solidify them faster than they do while we are actually playing. And we also need to train the brain's retrieval mechanism, as we are not just learning movements, but recalling them in real time. Understanding these things has changed a lot about how I approach practice.

I started working with much shorter chunks, targeting 2 to 5 measures instead of running whole pieces. I put a strict time limit on how long I could spend on any given section. I started randomizing the material. Scrambling my practice stack and

going from one piece to the next with a timer started showing results almost immediately! And I started taking breaks. Lots of short breaks. A 30-second break every few minutes works wonders, as do longer breaks. 20 minutes to several hours between sessions. What to do during them? Drinking water, stretching, exercise, and mental practice. All things I aspire to do and somehow manage to avoid. They're all important for their own sake, and much more conducive to learning than a "quick" dopamine hit from my phone. Implementing these ideas in my own practice and my students' sessions has accelerated my learning and given me back countless hours of time and helped avoid wear and tear on the body.

What was most difficult to implement was the shuffling. How do you arrange your stack of music to easily flow from one random passage to the next, sometimes recurring, sometimes not? And practicing different passages on the same page can be tricky if you're trying to interleave them all. Plus these days, my music is on a mix of paper and screen, adding to the difficulty of organizing practice. I decided I needed something to help me interleave more efficiently and came up with this app, Interleaves.

This app was originally intended just for me, so I built it around my particular needs and desires, leaving out things that I don't find helpful. As I started using it, I realized just how effective it is and decided to share it with friends and students. The response has been very positive. People are using it and finding real benefit, so I'm releasing it to the world :-)

<http://stuartzobel.com/interleave>

Upload your music: PDFs, images, or take a photo right in the app. Mark the passages you want to practice, set a timer and a randomization strategy, and hit Practice. The app shuffles your queue and keeps you moving. Set up rest breaks and it'll remind you to stretch, breathe, drink water or whatever you want it to. Your scores stay until you delete them, and you can export and reload collections when you want to switch focus.

Please give it a try. I think you will be surprised with the results. I would love to hear your thoughts on it, and if you like it, please pass it on to a friend!